PRACTICAL 8

Aim: Persistent Cross-Site Scripting Attack

A. Set up a vulnerable web application that is susceptible to persistent XSS attacks.

B. Craft a malicious script to exploit the XSS vulnerability and execute arbitrary code.

C. Observe the consequences of the attack and understand the potential risks associated with XSS vulnerabilities.

Theory: -

a. Persistent Cross-Site Scripting (XSS) is a type of security vulnerability commonly found in web applications.

b. It occurs when an attacker injects malicious scripts into a web application, which are then stored and executed whenever other users interact with the affected page.

1. Identifying XSS Vulnerabilities:

XSS vulnerabilities occur when user input is reflected back to users without proper encoding or validation.

These vulnerabilities can be identified by examining areas in the web application where user input is displayed without adequate sanitization.

Common vulnerable areas include input fields, comment sections, or any part of the application where user-generated content is rendered.

2. Crafting Malicious Scripts:

Malicious scripts are crafted by attackers to exploit identified XSS vulnerabilities.

These scripts are designed to perform harmful actions such as stealing cookies, redirecting users to malicious websites, or defacing the website.

Crafted scripts are typically injected into vulnerable input fields or areas of the web application.

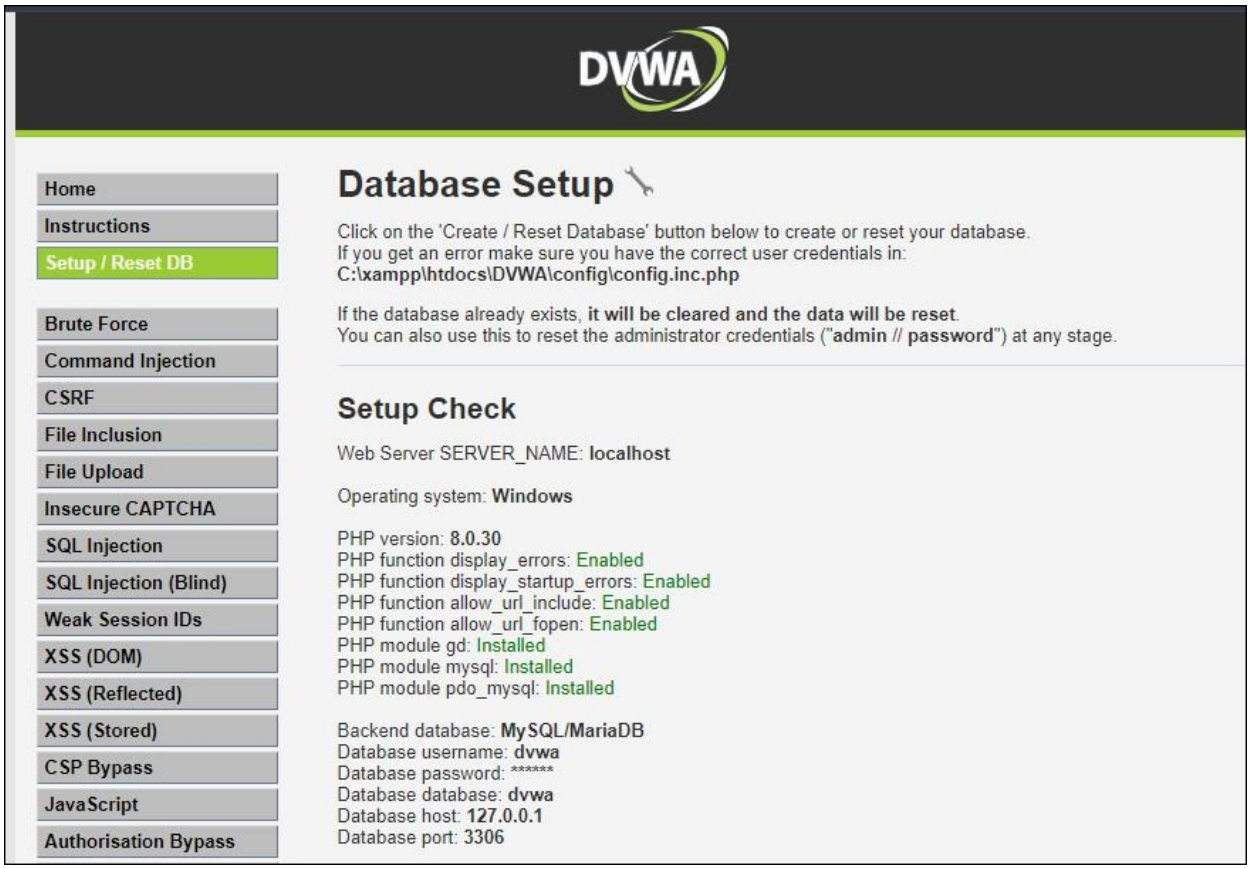
3. Exploiting the Vulnerability:

Once a malicious script is injected into the vulnerable input field or area, it becomes stored in the application's database.

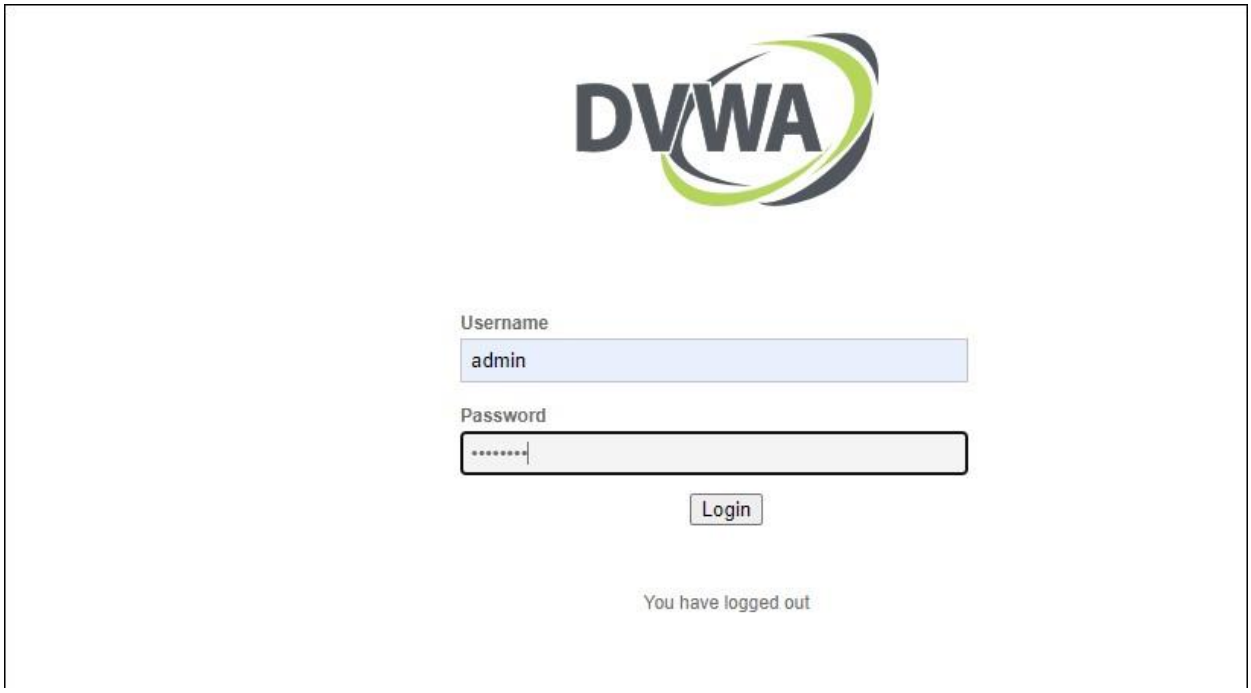
Whenever other users access the affected page, the injected script is executed within their browser.

This execution of malicious code can lead to various consequences, including session hijacking, phishing attacks, or the installation of malware.

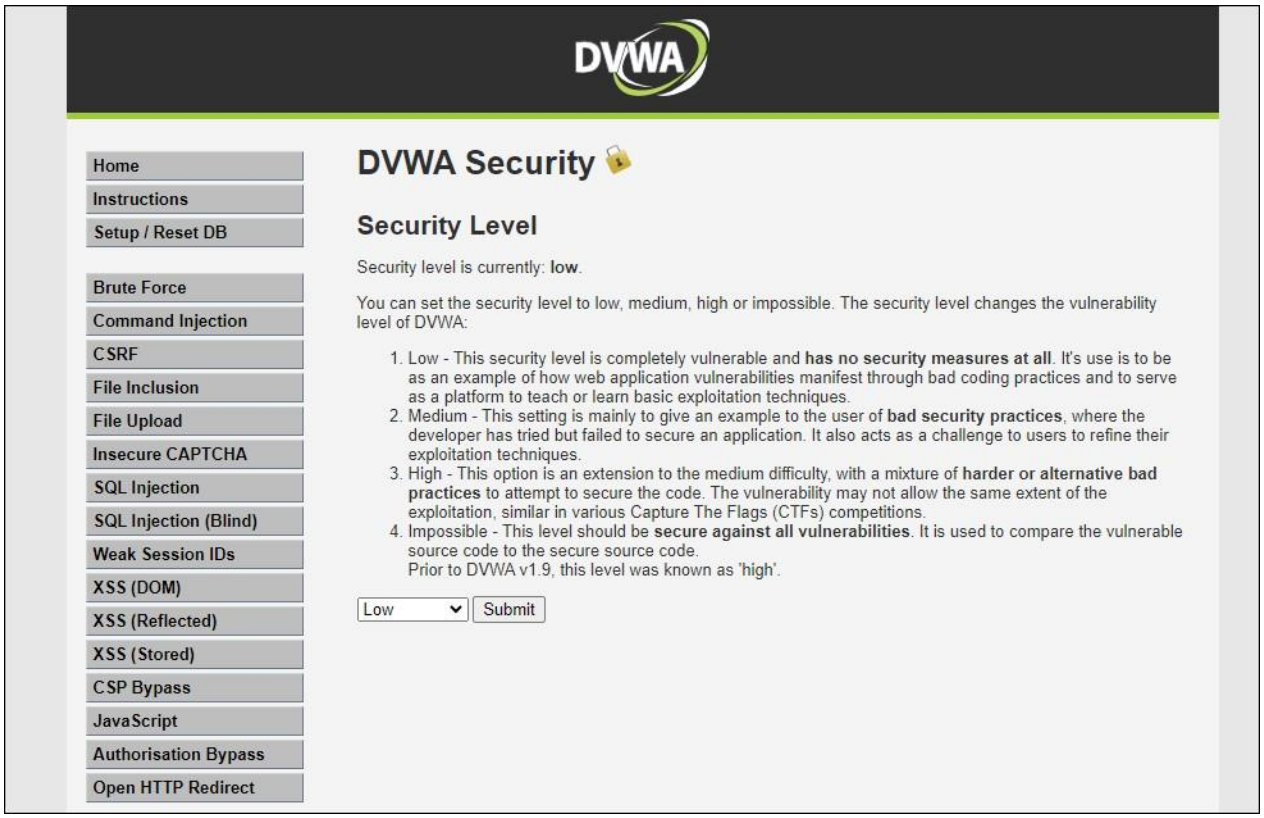
Step 1: - Click on Create/Reset Database.



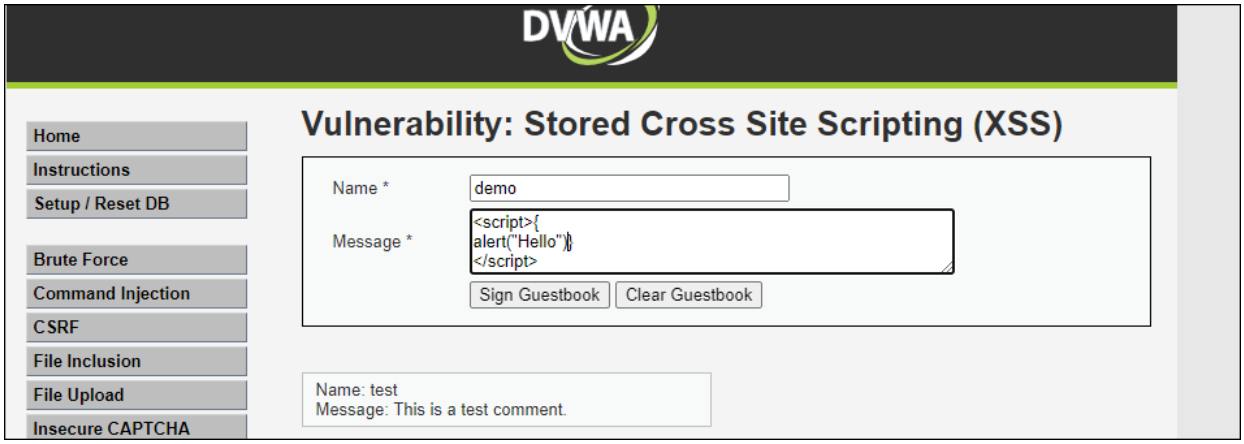
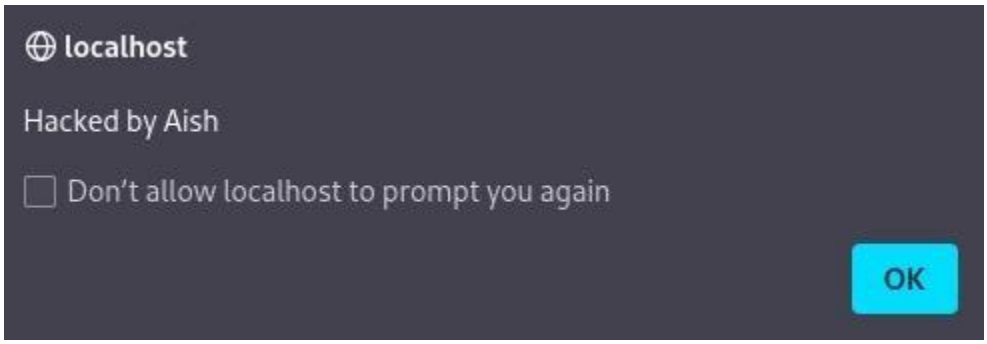
Step 2: - Login with Username and Password



Step 3: - Set the DVWA security level in DVWA security tab to low



Step 4: - Click on XSS (stored) -> Fill the Name and write script code in Message as Follows:



Step 5: - Click on Sign Guestbook. (using kali Linux)